

Project Name: STEMConstellation Academy Management and Learning Platform

Team no.: 1

	Team member Name
1	Sergio Aupart (Team Leader)
2	Ophilia Fradarick
3	Nana Akua Adjedu
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Submitted to: Dr. Mahmood Al-Humaimidi

Douglas College – CSIS Department
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Project Proposal

1. Diagnose Problem

STEMConstellation Academy is a growing educational organization offering STEM-based programs for children aged 5 to 15. As enrollment increases, the absence of a centralized management system has become a significant operational challenge. At present, the academy relies on spreadsheets, email exchanges, and paper-based methods to handle class scheduling, student registration, attendance tracking, and homework monitoring. This manual approach demands considerable time and effort from administrators and teachers, and becomes increasingly unmanageable as the academy continues to expand.

The consequences of these inefficiencies extend to all stakeholders. Teachers must communicate homework details and class updates through individual emails, while administrators manually maintain registration lists, waitlists, and student progress records. Students and parents lack a unified platform to view schedules, track academic progress, or access course information. This fragmented system reduces overall productivity, creates communication gaps, and may negatively impact the quality of the learning experience and customer satisfaction.

2. Prescribe Treatment

The proposed solution is a centralized, web-based Academy Management and Learning Platform accessible across desktops, tablets, and mobile devices. The platform will consolidate all core academy operations into a single system with role-based access for students, teachers, and administrators — eliminating the reliance on disconnected tools and manual processes.

Parents will be able to log in securely, browse available courses, register their children, view upcoming class schedules, track their child's academic progress, request session cancellations, and earn reward points redeemable for prizes. Teachers will have access to curriculum materials, their assigned schedules, and tools to submit class reports, assign homework, evaluate student performance, and communicate directly with administrators. The system will also allow teachers to set availability at the start of each semester and generate payment invoices.

Administrators will benefit from a comprehensive dashboard that centralizes student registrations, waitlist management, teacher scheduling, course configuration, revenue tracking, and task management. Automated email notifications will streamline communication between all parties, while a centralized database will ensure data is stored securely and is accessible at all times. The platform will replace the need for manual tracking and significantly reduce the administrative workload across the organization.

3. Describe Plan of Work

The project will begin with a requirements-gathering phase, during which the team will analyze the operational needs of the academy and define the core workflows for each user role. This phase will establish the foundation for all subsequent development decisions and ensure the system is aligned with real operational challenges faced by administrators, teachers, students, and parents.

Following requirements gathering, the team will move into system design, mapping out user interfaces and data flows for each module. Development will then proceed in stages, covering user authentication, course and schedule management, homework tracking, communication tools, and progress monitoring. Each module will be tested independently and then integrated, with a final round of usability testing to ensure the system is intuitive and functional for all user types.

Throughout the project, the team will maintain a project log documenting all meetings, individual contributions, and development milestones. The final deliverable will be a fully functional web-based management platform that addresses the academy's operational needs, improves communication across stakeholders, and supports the academy's continued growth.